

Lesson 05: Terraform Resources

Learning Objectives

- Understand Terraform resource blocks in detail
- Learn resource arguments, attributes, and meta-arguments
- Create S3 buckets with versioning, encryption, and public access settings
- Create DynamoDB tables with various configurations
- Create SQS queues with attributes
- Understand resource dependencies between services

1. Resource Block Anatomy

Every Terraform resource follows this structure:

```
resource "resource_type" "local_name" {
  argument1 = "value1"
  argument2 = "value2"

  # Nested configuration blocks
  nested_block {
    key = "value"
  }
}
```

Part	Meaning	Example
resource	Block type — declares a resource	—
"resource_type"	The type of resource to create	"aws_s3_bucket", "aws_dynamodb_table"
"local_name"	A name you choose to reference this resource within Terraform	"main", "data", "logs"
arguments	Configuration parameters for the resource	bucket, region, acl
attributes	Values exported by the resource after creation	id, arn, bucket_domain_name
nested blocks	Sub-configurations with their own arguments	attribute, logging, server_side_encryption_configuration

2. S3 Bucket — Arguments and Attributes

Basic Bucket

```
resource "aws_s3_bucket" "main" {  
  bucket = "my-tf-bucket-05"  
}
```

Bucket with Versioning

S3 versioning is configured as a separate resource in the AWS provider v5+:

```
resource "aws_s3_bucket" "main" {  
  bucket = "my-tf-bucket-05"  
}  
  
resource "aws_s3_bucket_versioning" "main" {  
  bucket = aws_s3_bucket.main.id  
  
  versioning_configuration {  
    status = "Enabled"  
  }  
}
```

Bucket with Encryption

```
resource "aws_s3_bucket" "main" {  
  bucket = "my-tf-bucket-05"  
}  
  
resource "aws_s3_bucket_server_side_encryption_configuration" "main" {  
  bucket = aws_s3_bucket.main.id  
  
  rule {  
    apply_server_side_encryption_by_default {  
      sse_algorithm = "AES256"  
    }  
  }  
}
```

Bucket with Public Access Block

```
resource "aws_s3_bucket" "main" {  
  bucket = "my-tf-bucket-05"  
}
```

```
resource "aws_s3_bucket_public_access_block" "main" {
  bucket = aws_s3_bucket.main.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls     = true
  restrict_public_buckets = true
}
```

Key Arguments for `aws_s3_bucket`

Argument	Description	Required
<code>bucket</code>	Name of the bucket	No (Terraform generates one if omitted)
<code>force_destroy</code>	Allow deletion of non-empty bucket	No (default <code>false</code>)
<code>tags</code>	Key-value map of tags	No

Key Attributes (read after creation)

Attribute	Description
<code>id</code>	Same as bucket name
<code>arn</code>	ARN: <code>arn:aws:s3:::bucket-name</code>
<code>bucket_domain_name</code>	DNS name: <code>bucket-name.s3.amazonaws.com</code>
<code>region</code>	Region where bucket was created

3. DynamoDB Table — Arguments and Attributes

```
resource "aws_dynamodb_table" "main" {
  name           = "TfTable"
  billing_mode   = "PAY_PER_REQUEST"
  hash_key      = "pk"

  attribute {
    name = "pk"
    type = "S"
  }

  # Optional: Global Secondary Index
  global_secondary_index {
    name           = "gsi_name"
    hash_key      = "gsi_pk"
    projection_type = "ALL"
  }
}
```

```
tags = {
  Environment = "learning"
}
```

Key Arguments for `aws_dynamodb_table`

Argument	Description	Required
<code>name</code>	Table name	Yes
<code>billing_mode</code>	"PAY_PER_REQUEST" or "PROVISIONED"	No (default "PROVISIONED")
<code>hash_key</code>	Partition key attribute name	Yes
<code>range_key</code>	Sort key attribute name	No
<code>attribute</code>	Attribute definitions (name + type)	Yes (must match keys)
<code>global_secondary_index</code>	Optional GSI configuration	No
<code>local_secondary_index</code>	Optional LSI configuration	No
<code>ttl</code>	Time-to-live settings	No
<code>tags</code>	Key-value tags	No

Attribute Types

Type	Code
String	"S"
Number	"N"
Binary	"B"

Key Attributes

Attribute	Description
<code>id</code>	Table name
<code>arn</code>	Full ARN
<code>stream_arn</code>	DynamoDB Stream ARN (if streams enabled)

4. SQS Queue — Arguments and Attributes

```
resource "aws_sqs_queue" "main" {
  name = "tf-queue-05"
```

```
# Optional attributes
delay_seconds          = 0
max_message_size       = 2048
message_retention_seconds = 86400
receive_wait_time_seconds = 0
visibility_timeout_seconds = 30
}
```

Key Arguments for `aws_sqs_queue`

Argument	Description	Default
<code>name</code>	Queue name	Required
<code>name_prefix</code>	Prefix for auto-generated name	No
<code>delay_seconds</code>	Delivery delay (0–900)	<code>0</code>
<code>max_message_size</code>	Max message size (1024–262144)	<code>262144</code> (256 KB)
<code>message_retention_seconds</code>	Retention (60–1209600)	<code>345600</code> (4 days)
<code>receive_wait_time_seconds</code>	Long-poll wait (0–20)	<code>0</code>
<code>visibility_timeout_seconds</code>	Visibility timeout (0–43200)	<code>30</code>
<code>fifo_queue</code>	Whether queue is FIFO	<code>false</code>
<code>tags</code>	Key-value tags	No

Key Attributes

Attribute	Description
<code>id</code>	Queue URL
<code>arn</code>	Full ARN
<code>url</code>	Queue URL (same as <code>id</code>)

5. Cross-Resource References

Resources can reference attributes of other resources using the `resource_type.local_name.attribute` syntax:

```
resource "aws_s3_bucket" "main" {
  bucket = "tf-bucket-references"
}

resource "aws_s3_bucket_versioning" "main" {
  bucket = aws_s3_bucket.main.id # <-- cross-reference
  versioning_configuration {
    status = "Enabled"
  }
}
```

```
}  
}
```

This creates an **implicit dependency**: Terraform knows it must create the S3 bucket before the versioning configuration.

6. Complete Example

```
# provider.tf  
provider "aws" {  
  access_key = "test"  
  secret_key = "test"  
  region     = "us-east-1"  
  
  endpoints {  
    s3          = "http://localhost:4566"  
    dynamodb   = "http://localhost:4566"  
    sqs         = "http://localhost:4566"  
  }  
  
  skip_credentials_validation = true  
  skip_requesting_account_id  = true  
  skip_metadata_api_check     = true  
  skip_region_validation      = true  
  s3_use_path_style           = true  
}
```

```
# main.tf  
resource "aws_s3_bucket" "data" {  
  bucket = "tf-resources-data-bucket"  
  tags = {  
    Name = "DataBucket"  
  }  
}
```

```
resource "aws_s3_bucket_versioning" "data" {  
  bucket = aws_s3_bucket.data.id  
  versioning_configuration {  
    status = "Enabled"  
  }  
}
```

```
resource "aws_dynamodb_table" "items" {  
  name           = "Items"  
  billing_mode   = "PAY_PER_REQUEST"  
  hash_key      = "item_id"  
  
  attribute {  
    name = "item_id"  
    type = "S"  
  }  
}
```

```
}

global_secondary_index {
  name          = "category-index"
  hash_key      = "category"
  projection_type = "ALL"
}

attribute {
  name = "category"
  type = "S"
}
}

resource "aws_sqs_queue" "orders" {
  name                  = "orders-queue"
  delay_seconds         = 0
  message_retention_seconds = 86400 # 1 day
}
```

7. Key Takeaways

- Each resource block has a **type**, a **local name**, **arguments** (inputs), and **attributes** (outputs)
- Sub-resources like `aws_s3_bucket_versioning` are separate resources that attach to a parent
- Cross-resource references (`aws_s3_bucket.main.id`) create implicit dependencies — Terraform orders creation automatically
- DynamoDB requires `attribute` blocks that match the `hash_key` and any `global_secondary_index` key names
- SQS queues have sensible defaults for most attributes — you only need to set `name`

example and exercise repo url

<https://github.com/eyu-se/terraform-training-material-with-projects>

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